

Data Analytics Internship Portfolio

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Track: Data Analytics Internship @ DecodeLabs

Week 2

Project 2: Exploratory Data Analysis (EDA) & Programmatic Profiling

Python

Pandas

Matplotlib

Seaborn

Building upon the sanitized dataset from Week 1, the objective shifted toward algorithmic data discovery. This phase utilized Python's analytical libraries to programmatically uncover underlying patterns, hidden correlations, and significant outliers within the enterprise data.

1. Statistical Summary & Central Tendencies

Utilizing Pandas' describe() framework, numerical matrices were profiled to calculate foundational metrics (Mean, Median, Standard Deviation, and Quartiles). This provided an immediate mathematical baseline of the dataset's geometry and distribution.

2. Unmasking Outliers (Anomaly Detection)

A statistical Boxplot visualization was generated using Seaborn to map the variability of the TotalPrice feature.

Finding: The plot successfully isolated significant data points beyond the upper whisker. In a business context, these are not "noise" or entry errors, but rather critical "signals" representing high-value VIP customer purchases or bulk B2B transactions that require dedicated retention strategies.

3. Correlation vs. Causation Analysis

A Pearson Correlation Coefficient was calculated programmatically to investigate the relationship between product cost and consumer purchasing volume.

• **Coefficient Result:** $r \approx 0.028$

Analytical Insight: *The result indicates near-zero linear correlation. Customers purchase the quantities they require irrespective of the unit price variance. This lack of dependency was further visually validated via a generated Scatter Plot.*

4. Categorical Performance & Business Recommendations

Through GroupBy aggregation on the Product axis, a Bar Chart was constructed to rank inventory demand. The visual evidence clearly identified the top-performing categories (e.g., Printers and Chairs), serving as a direct compass for inventory procurement forecasting and targeted marketing allocations.